

External longitudinal validation of metabolic syndrome risk scores for cardiovascular, diabetes, and all-cause mortality in the US NHANES Linked Mortality File

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Abstract

Importance. Two metabolic syndrome risk scoring methods, a continuous score based on triangular areal similarity (RMRS) and a decision-tree formulation, have been developed and internally validated on Korean cohorts. External validation in a US cohort against established clinical risk equations, with long-term mortality outcomes, has not been reported.

Objective. To externally validate the RMRS and the metabolic syndrome decision tree against the ACC/AHA Pooled Cohort Equations (PCE), Framingham 2008, and FINDRISC for cardiovascular, diabetes-related, and all-cause mortality in a representative US adult population.

Design, setting, and participants. Prospective cohort using NHANES survey cycles 1999 through 2018 linked to the public-use NHANES Linked Mortality File (2019 release). Adults aged 20 to 79 years with non-missing data on the fasting metabolic syndrome components were eligible. Cohort N and exclusions reported in the Results.

Exposures. Five risk scores computed at NHANES baseline: RMRS, the metabolic syndrome decision tree, PCE, Framingham 2008, and FINDRISC.

Main outcomes and measures. Cardiovascular mortality, diabetes-related mortality (broadened definition; see Methods), and all-cause mortality, ascertained via the Linked Mortality File. Discrimination was quantified by inverse-probability-of-censoring weighted (IPCW) time-dependent AUC at 5 years and at off-cap horizons (9.5 years for cardiovascular and all-cause; 14.5 years for diabetes-related), with a survey-weighted AUC reported as a sensitivity. Calibration, net benefit (decision curve analysis), pairwise delta-AUC, and incremental value (continuous NRI and IDI) were reported per the analysis plan. Confidence intervals were computed via a within-stratum PSU-cluster bootstrap at 500 resamples.

Results. The pooled fasting-subsample cohort comprised 17,031 adults aged 20 to 79 years, with 806 all-cause deaths within the 10-year follow-up cap. Cardiovascular and diabetes-related (cause-specific) analyses used the 13,836 participants from cycles with full leading-cause coding (1999 to 2014), among whom 173 cardiovascular deaths accrued within the 10-year cap and 77 diabetes-related deaths under the broadened definition within the 15-year cap. For cardiovascular and all-cause mortality at the 9.5-year horizon, the clinical risk equations clearly outperformed the metabolic-syndrome-derived scores; Framingham reached a bootstrap-estimated AUC of 0.858 (95% CI 0.839 to 0.877) for cardiovascular mortality and 0.810 (0.800 to 0.821) for all-cause mortality, compared with RMRS at 0.660 (0.633 to 0.690) and 0.600 (0.585 to 0.617). For diabetes-related mortality at the 14.5-year horizon, RMRS and FINDRISC were comparable (RMRS 0.752 [0.716 to 0.790] vs FINDRISC 0.770 [0.729 to 0.812]), with the decision tree close behind at 0.692 (0.647 to 0.732). The H3 non-inferiority test (RMRS vs FINDRISC for diabetes-related mortality, margin -0.05 AUC) gave a primary IPCW time-dependent delta-AUC of -0.020 (within-stratum bootstrap 95% CI -0.068 to $+0.028$), whose lower bound fell below the margin, so non-inferiority was not established on the primary metric (a secondary binary-outcome DeLong delta was $+0.013$ [-0.043 , $+0.063$]). Adding RMRS on top of FINDRISC for diabetes-related mortality produced an IDI of 0.004 (0.001 to 0.009) and a continuous NRI of 0.139 (0.047 to 0.267), both with cluster-bootstrap intervals excluding zero. An XGBoost ceiling on the five metabolic syndrome components alone reached 0.772 AUC for diabetes-related mortality at the 14.5-year horizon, marginally above RMRS and FINDRISC and above the decision tree; adding age and sex raised that ceiling to 0.828.

Conclusions and relevance. Metabolic-syndrome-derived risk scores do not substitute for established clinical risk equations when the outcome is cardiovascular or all-cause mortality. For diabetes-related mortality, FINDRISC was numerically higher than RMRS on the primary time-dependent AUC and non-inferiority of RMRS was not established at the -0.05 margin, so the two scores are best read as statistically indistinguishable rather than one being non-inferior or superior to the other. With the full published decision tree reconstructed and applied with its native Korean feature scaling, the tree was competitive with RMRS: statistically close on all-cause mortality and modestly behind on the cause-specific outcomes. An earlier impression that the continuous score dominates the tree reflected a degraded three-value reconstruction of the published model rather than a methodological gap.

Introduction

Metabolic syndrome is a clustering of central adiposity, dysglycemia, dyslipidemia, and elevated blood pressure that is associated with increased risk of cardiovascular disease, type 2 diabetes, and all-cause

mortality [NCEP Expert Panel, 2002, Alberti et al., 2009]. Several scoring frameworks have been proposed to quantify metabolic syndrome severity on a continuous scale or via decision-tree classification, with the aim of improving prevention beyond the binary NCEP ATP III or IDF criteria.

Two such frameworks have recently been developed and internally validated on Korean cohorts. The Robust Metabolic Syndrome Risk Score (RMRS) maps the five component features into a continuous risk index via triangular areal similarity [Shin et al., 2024]. A complementary decision-tree formulation segments the input space using the same five component features [Shin et al., 2023].

Continuous metabolic syndrome severity scoring has a longer track record outside this group. The most developed line is the metabolic syndrome severity Z-score of DeBoer and Gurka, which has been placed head-to-head against the Pooled Cohort Equations and the Framingham score in the Atherosclerosis Risk in Communities and Jackson Heart studies; in those comparisons the severity score added discrimination for incident type 2 diabetes but little for incident cardiovascular disease beyond the clinical equations [Guo et al., 2018, Gurka et al., 2018]. At the level of the syndrome itself rather than a derived score, meta-analyses place metabolic syndrome at roughly a 1.3 to 1.6 fold higher risk of all-cause mortality and around a two-fold higher cardiovascular risk in general populations [Galassi et al., 2006, Mottillo et al., 2010]. Two features of this prior work leave room for the present study: the head-to-head comparisons evaluated incident disease rather than mortality and did not include a diabetes-specific screening tool, and FINDRISC has been examined as a mortality predictor only in Finnish cohorts [Silventoinen et al., 2005], not in a representative US sample.

External validation of the RMRS and the decision tree in a non-Asian population, against established clinical prediction equations, on long-term mortality outcomes, has not been reported. The RMRS report did apply the score to a US NHANES cross-section, but without mortality linkage or longitudinal follow-up [Shin et al., 2024], so its behavior as a longitudinal mortality predictor in this population remains untested. The ACC/AHA Pooled Cohort Equations [Goff et al., 2014] and the Framingham 2008 general cardiovascular risk profile [D’Agostino et al., 2008] are the primary baselines for cardiovascular risk prediction in US clinical practice. FINDRISC [Lindstrom and Tuomilehto, 2003] is the most widely used non-invasive screening tool for incident type 2 diabetes. A head-to-head longitudinal comparison on a representative US cohort would clarify the place, if any, of metabolic-syndrome-derived scores within the existing prediction-tool ecosystem. A parallel line of work has developed and validated separate noninvasive metabolic syndrome models with cardiovascular risk assessment on multicohort data [Park et al., 2025]; that work builds new models rather than externally validating the specific Shin/Shim/Oh

scores examined here.

We performed this comparison using NHANES survey cycles 1999 through 2018 linked to the public-use Linked Mortality File (2019 release) [National Center for Health Statistics, 2019]. The analysis followed a written study protocol and analysis plan, included in the project repository, and follows TRIPOD+AI [Collins et al., 2024] and STROBE [von Elm et al., 2007] reporting standards, and reports discrimination, calibration, net benefit, pairwise delta-AUC, and incremental value for three mortality outcomes (cardiovascular, diabetes-related, all-cause) across all five risk scores. The protocol’s primary non-inferiority hypothesis (H3) tested whether RMRS is non-inferior to FINDRISC for predicting diabetes-related mortality at the 14.5-year horizon, with a margin of -0.05 AUC set in the analysis plan.

Methods

Study design and data sources

We used a prospective cohort design on the public-use NHANES survey [National Center for Health Statistics, 2019] cycles A through J (calendar years 1999 through 2018) linked to the National Center for Health Statistics public-use Linked Mortality File (LMF) 2019 release. Both files are de-identified, publicly available, and do not require IRB approval for secondary analysis. NHANES employs a complex multi-stage probability sample of the US civilian, non-institutionalized population with sampling stratification (`sdmvstra`), primary sampling units (`sdmvpsu`), and per-cycle survey weights. Because the cohort is the morning fasting subsample, the weighted analyses used the fasting-subsample weight (`WTSF2YR`) rather than the MEC examination weight, following NCHS guidance for analyses involving fasting-subsample measurements; weights were divided by the number of pooled two-year cycles, and participants with a zero fasting weight (not selected into the fasting subsample) were dropped from the weighted estimates. The weighted survey design (`R survey` package) was applied to the hazard models and the baseline table. The rank-based time-dependent AUC and the decision curve analysis are reported unweighted as the primary estimates, with a survey-weighted AUC provided as a sensitivity (see Results).

The LMF provides mortality follow-up time in months (`PERMTH_EXM`) and the NCHS 113-cause underlying cause-of-death recode (`UCOD_LEADING`), plus contributing-cause flags for diabetes and hypertension.

Participants and inclusion criteria

Eligible participants were adults aged 20 to 79 years at NHANES examination with non-missing values on the metabolic syndrome components (waist circumference, fasting glucose, HDL cholesterol, triglycerides, and blood pressure) and no prior cardiovascular disease or diagnosed diabetes. The 20 to 79 window matches the protocol eligibility criterion; PCE was computed only on its applicable 40 to 79 subset (reported separately). Because triglycerides and HDL are measured only in the morning fasting subsample, the cohort is restricted to fasting-subsample participants, which removes approximately half of adult NHANES respondents. The fasting-lab tables are published under legacy NHANES names for the 1999 to 2004 cycles and under the standardized names from 2005 onward; both were retrieved so that all ten cycles contribute. The HDL column name varies across cycles (LBDHDL in cycles A and B, 1999 to 2002; LBXHDD in cycle C, 2003 to 2004; LBDHDD in cycles D onward, 2005 to 2018) and was coalesced. Race and ethnicity were coalesced from RIDRETH3 (available from cycle G, 2011 onward) and RIDRETH1 (earlier cycles), with the limitation that the non-Hispanic Asian category is not separately identifiable before cycle G.

Risk scores

Five risk scores were computed at NHANES baseline.

RMRS [Shin et al., 2024] maps the five metabolic syndrome components to a continuous index via triangular areal similarity to a reference healthy point. Implementation follows the published formula without modification, using the published US waist thresholds.

Metabolic syndrome decision tree [Shin et al., 2023] classifies subjects through a published depth-5 CART. The complete tree (22 terminal nodes) was reconstructed from the paper’s Supporting Information S1 Table, which lists the full ruleset and the binary class at each leaf. The paper does not tabulate per-leaf probabilities; these were read from the published risk map (its Figure 12) and anchored to the paper’s worked example (its Figure 7, which routes a stated risk of 0.31 to one leaf), so the leaf risks are approximate to the resolution of that figure while the split structure and class labels are exact. Because the tree was trained on Korean-threshold-scaled waist and blood-pressure features, it was applied with the Korean scaling its splits expect; applying it with US waist thresholds would misalign the routing. A sensitivity analysis refitting a tree on a held-out NHANES split is reported in the Results.

ACC/AHA Pooled Cohort Equations (PCE) [Goff et al., 2014] estimate 10-year atherosclerotic car-

cardiovascular disease risk. The four sex- and race-specific equations were implemented from the original 2013 Goff et al. coefficients (the 2018 Yadlowsky revision is a separate model and was not used). PCE applies to adults aged 40 to 79 without prior atherosclerotic cardiovascular disease; subjects outside this range or with missing race were excluded from PCE-specific analyses (PCE-eligible N reported in Results). For the Hispanic, non-Hispanic Asian, and other race groups, the white-stratum equations were used as the default approximation, following the ACC/AHA risk-estimator convention.

Framingham 2008 general cardiovascular risk profile [D’Agostino et al., 2008] estimates 10-year general cardiovascular risk. Coefficients from the original publication were applied.

FINDRISC [Lindstrom and Tuomilehto, 2003] is an integer screening score (range 0 to 26) for incident type 2 diabetes risk. NHANES does not collect FINDRISC items directly; the implementation maps NHANES variables onto FINDRISC items as follows. Age, body mass index, and waist circumference map directly. Physical activity is derived from the physical activity questionnaire (PAQ): a positive response to any moderate or vigorous recreational or work activity item marks the participant as active, using PAD200/PAD320 for cycles A through D (1999 to 2006) and PAQ650/PAQ665/PAQ605/PAQ620 for cycles E through J (2007 to 2018), which span the 2007 PAQ redesign. Antihypertensive medication use maps from BPQ050A. Personal history of high blood glucose is taken as a positive prediabetes report (DIQ160, available cycles D onward) or a measured impaired fasting glucose of 100 to 125 mg/dL, the latter available in every cycle. Family history of diabetes maps from MCQ300C (available cycles D onward), mapped to the first-degree tier when positive. The daily vegetable and fruit item has no cycle-consistent NHANES counterpart and is held at the no-credit level; its omission is the one residual proxy limitation and is noted in the Discussion. MCQ300C and DIQ160 are absent before cycle D (1999 to 2004), so family history defaults to none and the high-blood-glucose item relies on the measured fasting-glucose component in those cycles.

Outcomes

Three mortality outcomes were defined.

Cardiovascular mortality was defined as a leading underlying cause of disease of the heart (UCOD_LEADING code 1) or cerebrovascular disease (code 5). The follow-up cap was 10 years; AUC was reported at $t = 5$ and at the off-cap horizon $t = 9.5$ years (AUC at exactly the cap is degenerate under IPCW). The 2015-2016 and 2017-2018 public-use files release only leading-cause codes 1, 2, and 10, folding cerebrovascular and other specific causes into “all other”, which we verified directly in the linked

data; cardiovascular and diabetes-related (cause-specific) analyses were therefore restricted to cycles A through H (1999 to 2014), which carry the full leading-cause coding. All-cause mortality used every cycle.

All-cause mortality was defined as any death recorded in the LMF. The follow-up cap and horizons matched the cardiovascular outcome, using the complete 1999 to 2018 cohort.

Diabetes-related mortality used a broadened definition: UCOD_LEADING for diabetes mellitus (code 7), OR UCOD_LEADING for nephritis (code 9), OR the LMF DIABETES contributing-cause flag equal to 1. The follow-up cap was extended to 15 years to accumulate adequate events under the broadened definition; AUC horizons were $t = 5$, $t = 10$, and $t = 14.5$. The protocol’s narrow definition (UCOD_LEADING = 7 only) yielded 6 events at 10 years and was underpowered for Fine-Gray modeling; the broadening to include nephritis and the contributing-cause flag, plus the follow-up extension, is documented as a deviation in the analysis plan.

Statistical analysis

For cardiovascular and diabetes-related mortality, we modeled cumulative incidence under competing risks using survey-weighted Fine-Gray subdistribution hazards [Fine and Gray, 1999], fit via the Geskus weighted-risk-set construction (`survival::finegray`) with the fasting-subsample weight (WTSAF2YR) carried into the subdistribution Cox model and Huber-White standard errors clustered on the primary sampling unit. Subdistribution hazard ratios are reported per 1 standard deviation of each score. For all-cause mortality, we used survey-weighted Cox proportional hazards. Time-dependent discrimination was quantified with IPCW AUC [Blanche et al., 2013] via the `timeROC` R package with `iid = FALSE` (the `iid = TRUE` setting allocates an $N \times N$ matrix that exceeds available memory at the realized cohort size). The time-dependent AUC is a rank-based discrimination metric and is reported unweighted as the primary estimate; a survey-weighted AUC on the horizon-cap binary outcome, using the fasting-subsample weight, is reported as a sensitivity to gauge the design effect on discrimination.

Calibration was assessed via IPCW Brier scores at each primary horizon (`riskRegression::Score`), each computed on the score’s recalibrated horizon-specific absolute risk rather than the raw score, since several scores are not on a probability scale; this uses the same per-score recalibration as the decision curve analysis and is therefore apparent in-sample. Net benefit was reported via decision curve analysis [Vickers and Elkin, 2006] over a threshold range of 1% to 30%. For the competing-risks outcomes (cardiovascular, diabetes-related), each score was recalibrated to a horizon-specific absolute risk via the

Fine-Gray cumulative incidence function, and net benefit was computed with the Aalen-Johansen cause-specific incidence within the treat-positive group so that censoring and competing deaths are handled rather than ignored; for all-cause mortality the recalibration used a Cox model and the Kaplan-Meier complement. Decision thresholds for the single-threshold bootstrap summaries were placed on each outcome’s achievable predicted-risk range (5% for all-cause, 2% for cardiovascular, 1% for diabetes-related), since cardiovascular- and diabetes-mortality predicted risks do not reach the higher thresholds used for composite cardiovascular event scores. Each score’s recalibration was fit on the analysis cohort and evaluated on the same cohort, so the reported net benefit reflects apparent in-sample performance; the decision curve analysis is unweighted and exploratory.

For the protocol pairwise comparisons (H1: RMRS vs B9 on each outcome; H2: PCE vs Framingham on cardiovascular; H3: RMRS vs FINDRISC on diabetes-related), the point delta-AUC and its bootstrap confidence interval were taken from the paired IPCW `timeROC` fit on the common non-missing subsample. Definitive confidence intervals for all primary discrimination and net-benefit quantities were computed via a PSU-cluster bootstrap at 500 resamples that respects the survey design by resampling primary sampling units within each design stratum (`sdmvstrata`). A DeLong test on the horizon-cap binary outcome (`pROC::roc.test`, excluding subjects censored before the horizon) is reported only as a secondary check, since it ignores both the survey weights and the PSU clustering, uses a different (binary) outcome encoding than the primary time-dependent metric, and is therefore not the basis for any conclusion. The IPCW `timeROC` delta-AUC and its within-stratum bootstrap interval are the primary inference for the H3 non-inferiority test against the -0.05 AUC margin.

Incremental value of adding RMRS on top of FINDRISC for diabetes-related mortality was reported as continuous NRI [Pencina et al., 2008] and IDI. Point estimates were computed via `survIDINRI::IDI.INF` on a cause-specific hazard model with competing deaths censored, the standard input for this estimator; the 95% confidence intervals were taken from the same within-stratum PSU-cluster bootstrap so that the reclassification intervals respect the survey design, rather than the i.i.d. perturbation the estimator uses internally.

The analysis was complete-case on the metabolic syndrome components that define eligibility. Scores requiring additional inputs were set to missing where those inputs were unavailable (for example, PCE and Framingham require total cholesterol), and the affected score’s analyses were run on its non-missing subset; the per-score analytic N is reported with each result. The analysis plan named a Bonferroni adjustment for the five primary tests and a Benjamini-Hochberg adjustment for the secondary

set; the primary inference here is interval-based (bootstrap confidence intervals against the protocol margin and against zero), so no p-value adjustment is applied, and any DeLong p-values are reported only as unadjusted secondary checks.

Sensitivity analyses

The following sensitivity analyses were specified in the analysis plan (section 8) and are reported below.

Survey-weighted discrimination. A survey-weighted time-dependent AUC, computed on the horizon-cap binary outcome with the fasting-subsample weight, is compared with the unweighted primary AUC for every score and outcome (Table 3), to quantify how much the NHANES survey design shifts the discrimination estimates.

ML upper-bound benchmark. An XGBoost classifier with $\text{PSU} \times \text{stratum}$ -grouped 5-fold cross-validation (median imputation fit on each training fold) was used as a non-parametric ceiling for the achievable AUC. Two input sets were evaluated: the five metabolic syndrome components alone (waist circumference, systolic and diastolic blood pressure, HDL cholesterol, fasting glucose, triglycerides), which bounds what the metabolic syndrome signal itself supports, and the same components plus age and sex, an upper reference matching the demographic inputs the PCE and Framingham equations also use. Hyperparameters were not tuned beyond defaults. The cross-validated AUC is unweighted and uses a binary horizon-capped label, so it approximates rather than reproduces the IPCW time-dependent AUC.

Decision-tree refit. As an exploratory sensitivity analysis, a CART was refit on a single held-out 30% NHANES split, using the three BPWC features to predict ATP III metabolic syndrome, to separate the contribution of the original tree's Korean calibration from that of the tree method itself; results are in Table 5. This is an internal US comparison on one split, not an external mortality validation of a refit tree.

Subgroup analyses. Time-dependent AUC was computed within strata defined by sex, race/ethnicity, and age band (20 to 39, 40 to 49, 50 to 59, 60 to 69, 70 to 79 years), with strata of fewer than 10 events or 50 subjects suppressed. The non-Hispanic Asian category is reported only for cycles G onward; the diabetes-related strata are summarized in Table 6.

Reporting standards and study protocol

The study followed TRIPOD+AI [Collins et al., 2024] and STROBE [von Elm et al., 2007] reporting checklists; completed checklists are provided in the Supplement. The study protocol and analysis plan, including the primary hypotheses, the eligibility criteria, and the non-inferiority margin, are included in the project repository alongside the code. This is a transparent written protocol rather than a timestamped prospective registration; analysis-plan deviations triggered during the study are recorded in the protocol document.

Software

Analyses were conducted in R 4.3.3 on Ubuntu. Key packages included `survey` (NHANES design), `riskRegression` (Fine-Gray subdistribution model and cumulative incidence prediction), `survival` (weighted-risk-set Fine-Gray, Cox, Aalen-Johansen incidence), `timeROC` (IPCW AUC), `pROC` (secondary DeLong delta-AUC), and `survIDINRI` (NRI and IDI point estimates). Decision curve net benefit was computed directly from the recalibrated risks and the Aalen-Johansen incidence. The XGBoost sensitivity used Python 3.11 with `xgboost` and `scikit-learn`. The R environment is pinned with `renv` and the Python environment with `uv`; all code is available at the public repository listed in Data and Code Availability. With these pinned environments, a clean checkout regenerates the analytic cohort and the result tables via the Makefile; the result tables are versioned in the repository, and the large bootstrap distribution files are regenerated by the bootstrap target rather than stored.

Results

Cohort

The pooled NHANES 1999 to 2018 fasting subsample after applying the inclusion criteria comprised 17,031 adults aged 20 to 79 years. Of these, 9,815 were within the PCE age applicability window (40 to 79 years) and contributed to PCE-specific analyses. Within the 10-year follow-up cap, 806 all-cause deaths accrued in the full cohort. The cause-specific cardiovascular and diabetes-related analyses used the 13,836 participants from the 1999 to 2014 cycles with full leading-cause coding (7,904 of them PCE-eligible); within this subcohort, 173 cardiovascular deaths accrued within the 10-year cap, and 77 diabetes-related deaths met the broadened definition (`UCOD_LEADING` for diabetes mellitus

or nephritis, or the LMF DIABETES contributing-cause flag) within the 15-year cap. Survey-weighted baseline characteristics, overall and by all-cause mortality status, are shown in Table 1. Decedents were older (weighted mean 58.8 versus 42.7 years), more often current smokers (35.7% versus 20.6%), and more often on antihypertensive treatment (31.7% versus 14.7%) than survivors.

Table 1: Survey-weighted baseline characteristics of the pooled NHANES 1999–2018 fasting subsample (N = 17,031), overall and stratified by all-cause mortality status within the 10-year follow-up cap. Continuous variables are reported as survey-weighted mean (standard deviation); categorical variables as weighted percentage. Counts are unweighted.

Characteristic	Overall (n = 17,031)	Alive (n = 16,225)	Death (n = 806)
Age, years	43.3 (15.0)	42.7 (14.7)	58.8 (14.8)
Body mass index, kg/m ²	28.3 (6.5)	28.3 (6.5)	28.5 (6.8)
Waist circumference, cm	96.9 (15.9)	96.7 (15.9)	101.4 (16.9)
Systolic blood pressure, mmHg	119.7 (15.9)	119.4 (15.5)	129.6 (21.1)
Diastolic blood pressure, mmHg	71.0 (10.8)	71.0 (10.7)	70.8 (12.9)
Fasting glucose, mg/dL	99.1 (16.9)	98.8 (16.6)	105.3 (22.2)
HDL cholesterol, mg/dL	54.1 (16.0)	54.1 (15.8)	53.6 (21.4)
Triglycerides, mg/dL	126.7 (110.2)	126.1 (110.8)	142.6 (91.9)
Total cholesterol, mg/dL	196.8 (40.9)	196.7 (40.8)	200.4 (42.6)
Female sex	51.7	52.1	40.1
Non-Hispanic White	68.3	68.0	77.1
Non-Hispanic Black	10.6	10.6	11.2
Mexican American	8.6	8.7	3.8
Non-Hispanic Asian	2.4	2.4	0.6
Current smoker	21.2	20.6	35.7
Antihypertensive treatment	15.3	14.7	31.7
Physically active	73.1	73.7	56.6
Family history of diabetes	26.2	26.3	22.9
Prior high glucose / prediabetes	38.8	38.6	43.8

Discrimination

Time-dependent AUCs at the primary horizons are summarized in Table 2. For all-cause mortality at the 9.5-year horizon, discrimination ranged from 0.585 (B9 decision tree) to 0.810 (Framingham 2008). For cardiovascular mortality at the same horizon, the range was 0.631 (B9) to 0.858 (Framingham). For diabetes-related mortality at the 14.5-year horizon, the range was 0.692 (B9) to 0.770 (FINDRISC), with RMRS at 0.752 between the decision tree and FINDRISC.

Table 2: Time-dependent AUC at the primary horizons with 95% within-stratum PSU-cluster bootstrap CIs (500 resamples). All-cause AUCs use the full cohort (N = 17,031; FINDRISC N = 16,996 after listwise deletion on FINDRISC inputs). Cardiovascular and diabetes-related AUCs use the 1999 to 2014 cause-coded subcohort (N = 13,836; FINDRISC diabetes N = 13,806). PCE AUCs are computed on the PCE-eligible subset within each cohort (all-cause N = 9,815; cardiovascular N = 7,904).

Score	Outcome	Horizon	AUC (95% CI)
Framingham 2008	All-cause	9.5y	0.810 (0.800, 0.821)
PCE	All-cause	9.5y	0.758 (0.746, 0.772)
FINDRISC	All-cause	9.5y	0.672 (0.659, 0.687)
RMRS	All-cause	9.5y	0.600 (0.585, 0.617)
B9 tree	All-cause	9.5y	0.585 (0.572, 0.599)
Framingham 2008	Cardiovascular	9.5y	0.858 (0.839, 0.877)
PCE	Cardiovascular	9.5y	0.810 (0.785, 0.838)
RMRS	Cardiovascular	9.5y	0.660 (0.633, 0.690)
B9 tree	Cardiovascular	9.5y	0.631 (0.598, 0.661)
FINDRISC	Diabetes-related	14.5y	0.770 (0.729, 0.812)
RMRS	Diabetes-related	14.5y	0.752 (0.716, 0.790)
B9 tree	Diabetes-related	14.5y	0.692 (0.647, 0.732)

Survey-weighted AUC sensitivity

Survey-weighted AUCs, computed on the horizon-cap binary outcome with the fasting-subsample weight, tracked the unweighted primary estimates closely (Table 3), with differences within about 0.04 AUC for every score and outcome, so the survey design does not materially change the discrimination ordering. For diabetes-related mortality the weighted estimates placed RMRS marginally above FINDRISC (0.790 vs 0.750), the reverse of the unweighted ordering and well within the overlapping intervals, reinforcing that the two scores are statistically indistinguishable on this outcome.

Table 3: Survey-weighted versus unweighted time-dependent AUC (95% within-stratum PSU-cluster bootstrap CIs). Weighted AUC uses the fasting-subsample weight on the horizon-cap binary outcome; the unweighted column is the primary IPCW estimate.

Score	Outcome	Horizon	Weighted AUC (95% CI)	Unweighted AUC
RMRS	All-cause	9.5y	0.616 (0.598, 0.636)	0.600
B9 tree	All-cause	9.5y	0.596 (0.577, 0.614)	0.585
Framingham 2008	All-cause	9.5y	0.796 (0.780, 0.814)	0.810
RMRS	Cardiovascular	9.5y	0.642 (0.604, 0.682)	0.660
Framingham 2008	Cardiovascular	9.5y	0.842 (0.814, 0.871)	0.858
RMRS	Diabetes-related	14.5y	0.790 (0.744, 0.832)	0.752
FINDRISC	Diabetes-related	14.5y	0.750 (0.702, 0.793)	0.770
B9 tree	Diabetes-related	14.5y	0.710 (0.658, 0.757)	0.692

Calibration

IPCW Brier scores at each primary horizon, computed on the recalibrated absolute-risk scale, are provided in the Supplement. The metabolic-syndrome-derived scores are miscalibrated when read as raw probabilities (RMRS is bounded in a narrow band and FINDRISC is an integer count rather than a probability), so Brier is reported on the per-score recalibrated risk, the same absolute-risk scale used for the decision curve analysis below; Brier on the raw scores is not meaningful and is not reported. Figure 1 shows predicted-versus-observed calibration on this recalibrated scale, with subjects binned by predicted risk and observed cumulative incidence estimated by Aalen-Johansen (cardiovascular and diabetes-related) or one minus Kaplan-Meier (all-cause). After recalibration the continuous scores track the diagonal across the risk range; the B9 decision tree maps subjects onto its distinct leaf-risk levels, so its calibration appears as a small number of binned points.

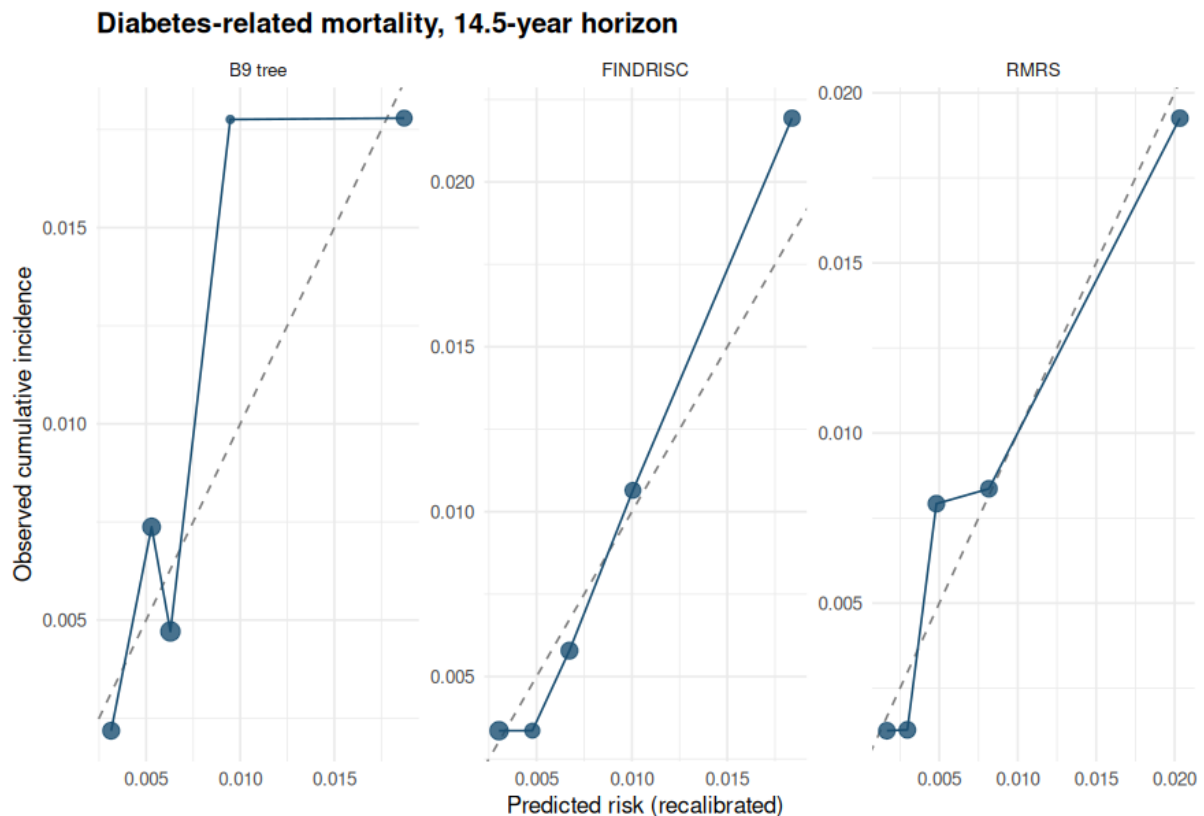


Figure 1: Calibration of recalibrated diabetes-related mortality risk at 14.5 years for RMRS, the B9 tree, and FINDRISC. Points are risk-bin means sized by bin count; the dashed line is perfect calibration. Calibration plots for all-cause and cardiovascular mortality are provided in the Supplement (results/plots/calibration_allcause.png, calibration_cv.png).

Net benefit

Decision curve analysis figures for the three outcomes are provided as `results/dca_allcause.png`, `results/dca_cv.png`, and `results/dca_dm.png`. For all-cause mortality at the 9.5-year horizon and a 5% decision threshold, PCE produced the largest net benefit (0.054 per person, 95% CI 0.050 to 0.058), followed by Framingham (0.030, 0.027 to 0.033), FINDRISC (0.018, 0.015 to 0.021), and RMRS and the B9 tree (both 0.012, 0.009 to 0.016 and 0.010 to 0.016 respectively). For cardiovascular mortality, predicted mortality risk does not reach the ACC/AHA 7.5% composite-event statin threshold, so net benefit was evaluated at 2%; PCE (0.010, 0.008 to 0.012) and Framingham (0.005, 0.004 to 0.007) led the metabolic-syndrome-derived scores, consistent with the AUC ordering. For diabetes-related mortality at a 1% threshold, FINDRISC, RMRS, and the B9 tree all gave small positive net benefit above treat-none (0.002 each, intervals excluding zero).

Pairwise comparisons and incremental value

The five protocol pairwise delta-AUC comparisons are summarized in Table 4.

Table 4: Protocol pairwise delta-AUC comparisons, computed as the paired IPCW time-dependent delta-AUC on the common non-missing subsample (the primary discrimination metric). CIs are 95% within-stratum PSU-cluster bootstrap percentile intervals (500 resamples). A positive delta favors the first score.

Contrast	Outcome (horizon)	Delta-AUC (95% CI)
RMRS vs FINDRISC (H3, non-inferiority)	Diabetes-related (14.5y)	-0.020 (-0.068, +0.028)
RMRS vs B9	All-cause (9.5y)	+0.014 (+0.000, +0.029)
RMRS vs B9	Cardiovascular (9.5y)	+0.029 (-0.003, +0.064)
RMRS vs B9	Diabetes-related (14.5y)	+0.060 (+0.015, +0.110)
PCE vs Framingham (H2)	Cardiovascular (9.5y)	+0.009 (-0.001, +0.021)

For the H3 non-inferiority test of RMRS versus FINDRISC on diabetes-related mortality at the 14.5-year horizon, the primary metric is the paired IPCW time-dependent delta-AUC. Its within-stratum bootstrap 95% percentile interval was (-0.068, +0.028), with a point estimate of -0.020. The lower bound fell below the protocol -0.05 margin, so on the primary metric non-inferiority was not established. This is consistent with the discrimination table, where FINDRISC has the higher IPCW AUC on this outcome (0.770 vs 0.752). A secondary DeLong delta on the horizon-cap binary outcome, which uses a different outcome encoding and ignores the survey design, was +0.013 (-0.043, +0.063) and would have cleared the margin, but it is reported only as a secondary check and is not the basis for the conclusion. The data are consistent with RMRS performing slightly below FINDRISC on this out-

come, and the two are best read as statistically indistinguishable rather than one being established as non-inferior to the other.

Adding RMRS on top of FINDRISC for diabetes-related mortality produced an IDI of 0.004 (95% CI 0.001 to 0.009) and a continuous NRI of 0.139 (95% CI 0.047 to 0.267), with both cluster-bootstrap intervals excluding zero. RMRS therefore captured a small amount of diabetes-mortality information beyond the FINDRISC items, of modest magnitude.

For the H2 comparison of PCE versus Framingham on cardiovascular mortality, the IPCW delta-AUC was +0.009 (95% CI -0.001 to +0.021) favoring PCE. The interval contained zero, indicating no detectable difference between the two clinical equations on this outcome in this cohort.

Sensitivity analyses

The XGBoost ceiling on the five metabolic syndrome components plus age and sex, with $\text{PSU} \times \text{stratum}$ -grouped 5-fold cross-validation, yielded AUCs of 0.813 for all-cause mortality at 9.5 years, 0.817 for cardiovascular mortality at 9.5 years, and 0.828 for diabetes-related mortality at 14.5 years. For all-cause mortality, the ML ceiling coincided with the Framingham AUC, indicating no headroom for a tree-based metabolic syndrome variant on this outcome. For cardiovascular mortality, the clinical Framingham equation at 0.858 exceeded the ML ceiling at 0.817, indicating that the metabolic syndrome inputs alone cannot reach the Framingham performance level even under flexible non-parametric modeling. For diabetes-related mortality, the ML ceiling at 0.828 exceeded RMRS at 0.752, FINDRISC at 0.770, and the B9 tree at 0.692, leaving roughly 6 to 8 AUC points of headroom for a tree-based variant of the metabolic syndrome representation specifically for diabetes mortality.

The full-tree B9 reconstruction was evaluated against a CART refit on US data to separate the effect of the tree's Korean calibration from the tree method itself. On a held-out 30% NHANES split (Table 5), the transported tree already reached AUCs of 0.581 for all-cause, 0.603 for cardiovascular, and 0.764 for diabetes-related mortality, the diabetes-related value above RMRS at 0.735 on the same held-out subjects. Refitting a CART on the training split using the same three BPWC features raised the held-out AUC to 0.635, 0.694, and 0.830 respectively, gains concentrated in the cause-specific outcomes. The data-driven tree independently routed the low-BPWC region to a single low-risk leaf (8.4% metabolic syndrome prevalence), whereas the published tree subdivides that region into nine leaves. The remaining gap between the transported and refit trees reflects re-estimation of the split thresholds and leaf risks on US data, consistent with transportability of the Korean calibration rather than a limitation of the tree

method. Because this rests on a single held-out split and a metabolic-syndrome (not mortality) refit target, it is exploratory and suggestive rather than a confirmed external validation of the refit tree.

Table 5: Decision-tree refit sensitivity. Time-dependent AUC on a held-out 30% NHANES split for the full transported B9 tree, a CART refit on the training split using the same three BPWC features to predict ATP III metabolic syndrome, and the continuous RMRS for reference.

Outcome	Horizon	B9 transported	B9 US-refit	RMRS
All-cause	9.5y	0.581	0.635	0.582
Cardiovascular	9.5y	0.603	0.694	0.660
Diabetes-related	14.5y	0.764	0.830	0.735

The protocol-specified subgroup analyses by sex, race/ethnicity, and age band for the diabetes-related H3 contrast are shown in Table 6. RMRS and FINDRISC tracked each other within most estimable strata, with FINDRISC ahead among women and in the youngest and oldest estimable bands and RMRS ahead in the 50 to 59 and 60 to 69 age bands; no subgroup reversed the overall near-equivalence. Strata with fewer than 10 diabetes-related deaths, including the non-Hispanic Asian group (present only from the 2011 cycle onward) and the youngest age bands, were not estimable and are omitted. For cardiovascular mortality, Framingham retained the highest AUC in every estimable sex, race/ethnicity, and age stratum, consistent with the pooled result; the full grid across all three outcomes is in `results/subgroup_auc.csv`.

Table 6: Pre-registered subgroup time-dependent AUC for the diabetes-related mortality H3 contrast (RMRS versus FINDRISC) at 14.5 years, within strata of sex, race/ethnicity, and age band (1999 to 2014 cause-coded subcohort). Strata with fewer than 10 diabetes-related deaths are omitted because IPCW AUC is unstable below that count; the omitted strata include non-Hispanic Asian (available only from the 2011 cycle onward) and the youngest age bands. The full grid across all outcomes is in `results/subgroup_auc.csv`.

Stratum	Level	Events	RMRS AUC	FINDRISC AUC
Sex	Male	49	0.728	0.734
	Female	28	0.778	0.832
Race/ethnicity	Mexican American	21	0.707	0.744
	Non-Hispanic White	23	0.780	0.781
	Non-Hispanic Black	21	0.776	0.802
Age band	50-59	11	0.779	0.627
	60-69	26	0.676	0.622
	70-79	27	0.603	0.641

Discussion

In this external longitudinal validation of two metabolic syndrome risk scoring methods (RMRS and the B9 decision tree) against three established clinical risk equations (PCE, Framingham 2008, FINDRISC) on the US NHANES Linked Mortality File, the headline finding is outcome-specific. The clinical risk equations clearly outperformed the metabolic-syndrome-derived scores when the outcome was cardiovascular or all-cause mortality. For diabetes-related mortality, the metabolic-syndrome-derived RMRS was statistically indistinguishable from FINDRISC, with non-inferiority not established at the -0.05 margin, and the two metabolic syndrome scores (RMRS and the decision tree) were close to one another and to FINDRISC.

For diabetes-related mortality at the 14.5-year horizon, RMRS reached an AUC of 0.752 against FINDRISC at 0.770, with a primary IPCW time-dependent delta-AUC of -0.020 and a within-stratum bootstrap 95% percentile CI of $(-0.068, +0.028)$. The non-inferiority test at the -0.05 margin was not met on the primary metric: the lower bound fell below the margin. FINDRISC is numerically the stronger of the two on the primary AUC, so the appropriate reading is that RMRS is statistically indistinguishable from, but not established as non-inferior to, the dedicated diabetes screening tool. The survey-weighted sensitivity placed RMRS marginally above FINDRISC on this outcome, the opposite ordering, and a secondary binary-outcome DeLong delta was mildly positive ($+0.013$); these conflicting signs across metrics underline that the two are within sampling noise of each other. The IDI and continuous NRI for adding RMRS on top of FINDRISC both excluded zero under cluster-bootstrap inference, indicating that RMRS captures a small amount of diabetes-mortality information beyond the FINDRISC items. This comparison uses a FINDRISC implementation with NHANES-derived family history, pre-diabetes or impaired-fasting-glucose, and physical-activity items rather than constant placeholders, so the near-equivalence is not an artifact of a degraded comparator. RMRS thus occupies a complementary niche in diabetes-mortality prediction alongside FINDRISC, which remains the established screening tool.

For cardiovascular mortality at the 9.5-year horizon, Framingham 2008 reached an AUC of 0.858 and PCE 0.810, with an IPCW delta-AUC of $+0.009$ between them (95% CI -0.001 to $+0.021$). The metabolic-syndrome-derived RMRS at 0.660 sat about 20 AUC points below these clinical equations, and the decision curve analysis showed the same ordering. For all-cause mortality, Framingham again led at 0.810, with PCE at 0.758 and the metabolic-syndrome-derived scores in the 0.585 to 0.600 range. The XGBoost ceiling on the metabolic syndrome components alone did not approach the clinical

equations; even adding age and sex (0.813 for all-cause and 0.817 for cardiovascular) left cardiovascular discrimination below Framingham's 0.858. The metabolic syndrome features do not contain the cardiovascular-mortality and all-cause-mortality information that the clinical equations integrate, regardless of the modeling approach used on them.

The faithful full B9 decision tree was competitive with the continuous RMRS, not dominated by it. RMRS exceeded the tree by a modest delta-AUC of +0.017 for all-cause (95% CI +0.004 to +0.031), +0.032 for cardiovascular (+0.003 to +0.067), and +0.081 for diabetes-related mortality (+0.040 to +0.126); the two were statistically close on all-cause mortality and separated only modestly on the cause-specific outcomes. This is a much smaller gap than the +0.071 to +0.158 that an earlier three-value reconstruction of the tree had produced, and the held-out refit analysis (Table 5) shows the transported full tree already matching or exceeding RMRS on diabetes-related mortality. The practical lesson is that a decision tree published only as a figure and a class-labeled ruleset can be made to look far weaker than it is if its leaf probabilities and feature scaling are not reconstructed faithfully; the apparent superiority of the continuous score over the tree was an artifact of reconstruction, not a property of the methods.

Several limitations apply. First, the published decision tree does not provide a numeric per-leaf probability table, so its leaf risks were read from the published risk map and anchored to the single worked example; the split structure and class labels are exact, but the leaf risks are approximate, and a different reading of the figure could shift the tree's discrimination by a small amount. Second, the FINDRISC implementation maps most items to NHANES variables but two gaps remain: the daily vegetable and fruit item has no cycle-consistent NHANES counterpart and is held at the no-credit level, and the family-history and self-reported-prediabetes items are only available from cycle D (2005 onward), so earlier cycles default those items; the residual proxy imprecision biases FINDRISC conservatively. Third, the headline discrimination metrics are unweighted; the survey-weighted sensitivity shows the design effect is small (within about 0.04 AUC), but the primary AUCs are sample rather than population estimates. Fourth, the cause-specific cardiovascular and diabetes-related analyses are restricted to the 1999 to 2014 cycles, because the public-use Linked Mortality File releases only three leading-cause categories for the 2015 to 2018 cycles; all-cause mortality uses the full cohort. Fifth, the cohort is restricted to the morning fasting subsample of NHANES, which removes approximately half of adult respondents and may introduce selection on factors associated with willingness or ability to fast. Sixth, the diabetes-related mortality outcome uses a broadened definition (UCOD for diabetes mellitus or nephritis, or the LMF

diabetes contributing-cause flag) with 77 events; the protocol’s narrow definition was underpowered for Fine-Gray modeling, and the broadening is documented as a deviation in the analysis plan, but the modest event count still limits precision for the cause-specific contrasts. We evaluated the two scores from this specific line of work and did not include the metabolic syndrome severity Z-score, a more established continuous formulation benchmarked against the clinical equations for incident disease [Guo et al., 2018, Gurka et al., 2018]; extending the present mortality comparison to that score is a natural next step. Future work should evaluate the metabolic syndrome inputs combined with the clinical risk equation inputs in joint models, and validate the diabetes-mortality finding in non-US cohorts.

The clinical implication is that metabolic-syndrome-derived risk scores, whether the continuous RMRS or the decision tree, occupy a complementary niche in diabetes-mortality prediction alongside FINDRISC and do not substitute for established clinical risk equations for cardiovascular or all-cause mortality prediction.

Data and code availability

The NHANES survey files and the NHANES Linked Mortality File are public, de-identified data from the National Center for Health Statistics. All analysis code, with a pinned R (`renv`) and Python (`uv`) environment that regenerates the analytic cohort and the result tables from a clean checkout via the Makefile, is available at <https://github.com/jayhemnani9910/longitudinal-mets-validation> (MIT license for code; CC-BY for figures and text); the versioned result tables are under `results/`. The written study protocol and analysis plan are included in the repository under `prereg/`.

Conflicts of interest

The author reports no conflicts of interest.

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